

Fifth Semester		CSE-57		
#	Course Code	Course Title	CRH	CN H
1.	CSE-3521	Computer Architecture	3	3
2.	CSE-3523	Microprocessors, Microcontrollers and Embedded Systems	3	3
3.	CSE-3524	Microprocessors, Microcontrollers and Embedded Systems Lab	1	2
4.	CSE-3527	Compiler	3	3
5.	CSE-3528	Compiler Lab	1	2
6.	CSE-3529	Systems Analysis and Design	3	3
7.	CSE-3532	Tools and Technologies for Internet Programming	2	4
8.	EEE-2421	Electrical Drive and Instrumentation	2	2
9.	EEE-2422	Electrical Drive and Instrumentation Lab	1	2
10.	URED-3503	Introduction to Political Thoughts and Social Behavior	1	2



5 th Semester		
ISCED Code	Course Code	Course Title
0714	CSE-3521	Computer Architecture
Credit Hours: 3	Contact Hours: 3	Type: Core, Engineering
Prerequisite: CSE-2305 (Digital Logic Design)		
Co-requisite:		

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30Marks
	SEE: Semester End Examination		50 Marks

Objectives:

The objectives of this course are to

- To introduce the student with CPU, memory, I/O organization and peripherals.
- To introduce students to the contemporary design and implementation of microprocessors.
- To help the student to demonstrate an understanding of micro architecture concepts and features employed by contemporary microprocessors, such as pipelining, data forwarding, superscalar, branch predictors and also competence of basic skills in optimizing assembly programs.

Course Learning Outcomes (CLOs):

#	CLO Description	Weightage (%)
1.	Understand the basic organization of computer and different instruction formats and addressing modes.	20%
2.	Analyze and Evaluate the performance of computer	20%
3.	Analyze various issues related to memory hierarchy.	30%
4.	Evaluate various modes of data transfer between CPU and I/O devices	20%
5.	Illustrate various inter connection structures of multi processors.	10%

Mapping of CLO-PLO:

#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO1	Understand the basic organization of computer and different instruction formats and addressing modes.	PLO1	Cognitive/ Understand	Lecture, Discussion	Class test, Assignment, Exam
CLO2	Analyze and Evaluate the performance of computer.	PLO1	Cognitive/ Analyze/Evaluate	Lecture, discussion, Competitive Problem solving	Assignment Class test, Exam
CLO3	Analyze various issues related to memory hierarchy	PLO1	Cognitive/Apply	Lecture, Class discussion,	Class test, Assignment



#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
				Competitive Problem solving	Exam
CLO4	Evaluate various modes of data transfer between	PLO1	Cognitive/ Evaluate	Lecture, Class discussion, Competitive Problem solving	Class test, Assignment Exam
CLO5	Illustrate various inter connection structures of multi processors	PLO2	Cognitive/Understand	Lecture, Class discussion, Competitive Problem solving	Class test, Assignment Exam

Course Content:

Segment	Contents	Duration	CLOs
Section-A (Midterm Exam: 30 Marks)			
1	Computer architecture Basic, History, Different terminologies of computing device, Types of Computer Architecture	02	CLO1
2	Understanding Program Performance, Defining Performance, Measuring Performance, CPU Performance and its factor, Evaluating performance, MIPS as a performance Measure	02	CLO1
3	Instruction and data access methods, Instruction Set, Stored-Program Concept, Operations of the computer Hardware, Operands of the computer Hardware (Design Principles of Computer Hardware) Representation of Instructions in the Computer, Logical Operations, Instructions for decision making, MIPS Addressing for 32-Bit Immediate and Addresses	03	CLO2
4.	Arithmetic and logical operations, floating point operations, ALU design, Signed and Unsigned numbers, Number Conversion and representation, Arithmetic Operations and Representation. Matrix-chain multiplication and longest common subsequence problems as examples, Complexity analysis of the algorithms. Multiplication, Division and Floating point Hardwar	4	CLO3
5.	Understanding Program Performance, Defining Performance, Measuring Performance, CPU Performance and its factor, Evaluating performance, MIPS as a performance Measure Instruction and data access methods, Instruction Set, Stored-Program Concept, Operations of the computer Hardware, Operands of the computer Hardware (Design Principles of Computer Hardware)	2	CLO2
6	Representation of Instructions in the Computer, Logical Operations, Instructions for decision making, MIPS Addressing for 32-Bit Immediate and Addresses Arithmetic and logical operations, floating point operations, ALU	4	CLO4



	design, Signed and Unsigned numbers, Number Conversion and representation, Arithmetic Operations and Representation		
7	Matrix-chain multiplication and longest common subsequence problems as examples, Complexity analysis of the algorithms. Multiplication, Division and Floating point Hardwar	2	CLO4
Section-B (SEE: 50 Marks)			
8	The control unit (Single cycle Data path) : hardwired and micro programmed, Logic Design Convention, Clocking Methodology, Data path Basic, State Element of Data path Building a single Data path (R-Type, I-Type and J-type Instructions) Designing the main control unit (with control signals effect), Operation of the data path. The control unit (Multiple cycle Data path): Hardwired and micro programmed, Multi-cycle implementation Basic. Necessity of multi-cycle implementation, Details of Control Signal Breaking the Instruction Execution into Clock Cycles, Exceptions	05	CLO5
9	Pipelining: Overview of Pipelining, Pipelining Hazard, Pipeline Data Path (R-Type, I-Type and J-type Instructions) Pipeline Control, Exception, Exception Handling	04	CLO5
10	Memory organization: Introduction to memory, memory Hierarchy, Basic on Cache, Accessing a cache, Handling cache miss and writes, Flexible placement of cache block	02	CLO3
11	Introduction to virtual Memory, Virtual to physical, Handling page Fault TLB, Integrating Virtual Memory, TLBs and Caches	02	CLO1
12	I/O systems, channels, interrupts, DMA, I/O Devices basic, Disk Storage and dependability, RAID, Buses and other connections between processors, Memory and I/O Devices, Interfacing I/O devices with processors.	02	CLO1
		34	

Books :**Text Book :**

1. John Hennesy, David Patterson, "Computer Organization", 3rd edition, Morgan Kaufman publisher
ISBN: 1-55860-604-1

Reference Books :

1. William Stallings, "Computer Organization and Architecture designing for performance", 8th Edition, Library of Congress Cataloguing-in-Publication data on file
2. M. Morris Mano e, "Digital Logic and Computer Design", 3rd Edition, Pearson Education

Course Assessment Pattern (Theory courses):

Bloom's Category		Evaluations out of 100 marks			
		CIE (50 marks)			SEE (50marks)
Cognitive	Affective	Mid-	Assignment/	Attendance	Written Exam



learning	learning	term (30)	Class Test (10)	Marks (10)	(50)
Remember		5			5
Understand		10	5		15
Apply		15	5		30
Analyze					
Evaluation					
Create					
x	Responding	x	x	10	

Note: CIE=Continuous Internal Evaluation, SEE= Semester End Examination



5 th Semester		
ISCED Code	Course Code	Course Title
0612	CSE-3523	Microprocessor, microcontroller and embedded system
Credit Hours: 3	Contact Hours: 3	Type: Core, Engineering
Prerequisite: CSE-2323 (Digital Logic Design)		
Co-requisite: CSE-3524 (Microprocessors, Microcontrollers and Embedded Systems Lab)		

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30Marks
	SEE: Semester End Examination		50 Marks

Objectives:

Microprocessor is responsible for processing the tasks within a computer .This course equips the students with the basic features , operations and techniques of 8086 microprocessor , the basics of using microcontroller and the basic characteristics of embedded systems. This course also enables the students to learn the basic concepts of assembly language programming using emulator such as EMU 8086.

Course Learning Outcomes (CLOs):

#	CLO Description	Weightage (%)
1.	Understanding of the basic operations, architecture, organization, assembly language instructions of microprocessor, microcontroller and embedded systems	70
2.	Apply assembly language programming to solve problems.	20
3.	Develop the skill of designing various microcontroller -based systems according to practical applications.	10

Mapping of CLO-PLO:

#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO1	Understanding of the basic operations, architecture, organization, assembly language instructions of microprocessor, microcontroller and embedded systems	PLO1	Cognitive/ Understand	Lecture, Discussion	Class test, Assignment, Exam
CLO2	Apply assembly language programming to solve problems.	PLO2	Cognitive/Apply	Lecture, discussion, Competitive Problem solving	Assignment Class test, Exam



#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO3	Develop the skill of designing various microcontrollers -based systems according to practical applications.	PLO3	Cognitive/Apply	Lecture, Class discussion, Competitive Problem solving	Class test, Assignment Exam

Course Content:

Segment	Contents	Duration	CLOs
Section-A (Midterm Exam: 30 Marks)			
1	Introduction: Introductory Concept: Evaluation of microprocessor, Types of microprocessor, system bus, hardware of a microprocessor, memory-addressing technique.	02	CLO1
2	8086 Microprocessor: properties, architecture, registers, FLAGS register, physical address calculation, addressing modes, Addressing Techniques of 8086 Microprocessor.	06	CLO1
3	Instruction: Instruction set, Instruction format, Fetch-decode-Execution cycle, basic concepts of assembly language	04	CLO2
Section-B (SEE: 50 Marks)			
4	Interrupt System: Sources of interrupt, Types of interrupt, handling interrupt request, interrupt vector and table, 8259A priority interrupt controller, Daisy chain..	04	CLO1
5	An overview of Intel 80186, 80286, 80386, 80486 and Pentium microprocessor; Advanced microprocessor: Embedded microprocessor, Bit slice microprocessor, arithmetic processor, Multitasking, Itanium and Merced Microprocessor.	06	CLO1
6	I/O operation: Isolated and memory mapped I/O, 8255A Programmable peripheral Interface, DMA technique, I/O ports, I/O processor	02	CLO1
7	Micro controller: Architecture of 8051, Signals, Operational features, Memory and I/O addressing, Interrupts, Instruction set, Applications. Design of a particular microcontroller based architecture.	04	CLO3
8	Embedded Systems: Introduction to Embedded Systems, Characteristics, Components, Hardware-Software Partitioning, Specification and Modeling, Optimizing Design, SOC, and Application Domains.	02	CLO1
		30	

Books :**Text Book :**

1. "Bary B Brey", The INTEL Microprocessors 8th edition Prentice Hall 2009, ISBN 13: 978-0-13-502645-8
2. "Mohamed Ali Mazidi Janice Gillispie Mazidi" "The 8051 microcontroller and embedded



systems” 2nd edition Pearson 2005, ISBN 13: 978-0131194021

Reference Books :

1. Dr. M. Rafiquzzaman: Microprocessors & Microcomputer -Based System Design
2. T. Hanley: Microprocessor and microcomputer
3. John F. wakerly: Micro Computer architecture and programming
4. John P. Hayes: Compute architecture and organization
6. Douglas V.: Hall, Microprocessor and Interface.
7. Ramesh Gaonker: Microprocessor Interfacing.

Course Assessment Pattern (Theory courses):

Bloom's Category		Evaluations out of 100 marks			
		CIE (50 marks)			SEE (50marks)
Cognitive learning	Affective learning	Mid-term (30)	Assignment/ Class Test (10)	Attendance Marks (10)	Written Exam (50)
Remember		15	5		15
Understand		10	5		15
Apply		5			20
Analyze					
Evaluation					
Create					
x	Responding	x	x	10	

Note: CIE=Continuous Internal Evaluation, SEE= Semester End Examination

5 th Semester		
ISCED Code	Course Code	Course Title
0612	CSE-3524	Microprocessor, microcontroller and embedded system sessional
Credit Hours: 1	Contact Hours: 2	Type: Core, Engineering
Prerequisite: CSE-2324 (Digital Logic Design Lab)		
Co-requisite: CSE-3523 (Microprocessors, Microcontrollers and Embedded Systems)		

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30Marks
	SEE: Semester End Examination		50 Marks

Objectives:

- To know about the basics of 8086 microprocessor operation
- Input data in different addresses using 8086 MDA toolkit.
- Serial and parallel port connection
- Understanding the use of assembly language using the concepts of 8086 microprocessor and microcontroller
- Design a small project using Proteus (open ended project) based on the concept of microcontroller

Course Learning Outcomes (CLOs):



#	CLO Description	Weightage (%)
1.	Accumulating knowledge about the structure of 8086 microprocessor, its instruction set and interfacing technique.	40
2.	Develop the ability to use a suitable trainer kit to load and verify assembly language program in 8086 microprocessor.	40
3.	Develop the ability to design microcontroller-based system for the solution of real-world problems.	20

Mapping of CLO-PLO:

#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO1	Accumulating knowledge about the structure of 8086 microprocessor, its instruction set and interfacing technique.	PLO1	Cognitive/Apply	Lecture, Class discussion, Lab work, Note	Assignment, Class performance, Exam
CLO2	Develop the ability to use a suitable trainer kit to load and verify assembly language program in 8086 microprocessor.	PLO5	Cognitive/Apply	Lecture, Class discussion, Lab work, Note	Assignment, Class performance, Exam
CLO3	Develop the ability to design microcontroller-based system for the solution of real-world problems.	PLO11	Cognitive/ Apply	Class discussion, Lab work, Note	Assignment, Class performance, Exam

Lecture Plan:

Week	Activities	Topics	CLOs
1	Lecture, Problem solving	Familiarization with MDA-8086 microprocessor kit and its operation in "Machine Code" mode.	CLO1
2	Lecture, Problem solving	Introduce the pins of 8086 microprocessor and its internal working procedures.	CLO1
3	Lecture, Problem solving	To load the machine codes of a sample program to MDA-8086, execute it and verify the results.	CLO1
4	Lecture, Problem solving	Introduction of Serial monitor mode operation of MDA-8086 and verification of arithmetic operations.	CLO1
5	Lecture	Introduction and implementation of various logic operations in assembly language.	CLO2
6	Lab test 1		



Week	Activities	Topics	CLOs
7	Lecture, Problem solving	Introduction and implementation of program control instructions in assembly language	CLO2
8	Lecture, Problem solving	Implement various interrupt system of 8086 microprocessor.	CLO2
9	Lecture, Problem solving	Understanding the concept of complex engineering and open ended lab problems	CLO3
10	Lecture, Discussion	Implementation of open ended lab : Kitchen timer and alarm system	CLO3
11	Lecture	Implementation of open ended lab : 4 way traffic system using Rduino (Both with Proteus simulation and hardware).	CLO3
12	Lecture, Problem solving	Solving problem from the above contexts	CLO3
13	Lab test 2		
14	Problem solving	Design and develop your own open ended lab problem	CLO3
15	Viva	Viva Voce and problem solving	CLO3

Books :**Text Book :**

1. Win8086 MANUAL, User's manual ,Modas Eng. Co.,ltd.
2. Assembly language programming and organization of the IBM PC, Ytha YuCharles Marut

Reference Books :

6. Dandamudi, Sivarama , Introduction to Assembly Language Programming, Sivarma P. Dandamundi, ISBN 978-4757-2858-3.



5 th Semester		
ISCED Code	Course Code	Course Title
0613	CSE-3527	Compiler
Credit Hours: 3	Contact Hours: 3	Type: Core, Engineering
Prerequisite: CSE-2427 (Theory of Computation)		
Co-requisite:		

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30Marks
	SEE: Semester End Examination		50 Marks

Objectives:

This is a course to familiarize the students with knowledge and ideas of the construction of computer program translators called compilers. To take this course, students should be able to gain the skills of automata theory which need to design and implementation of the compiler. The competitive and standard programming ability with some mathematical maturity also will be expected; students should have some idea of computer language grammar and computing skills will also be helpful. After studying this course the students will have in-depth knowledge of the design and implementation of compilers maintaining the standard steps.

Course Learning Outcomes (CLOs):

#	CLO Description	Weightage (%)
1.	Understand the knowledge of language translation to analyze the source program for compiler construction with its front-end and back-end phases.	30
2.	Apply the theory of computing (patterns, tokens, regular expressions, grammar and finite automata) to design and create the compiler for different programming languages and construction of compiler to improve language translation problem solving skills to basic real world situations	30
3.	Apply the Intermediate code generation tools to design the intermediate code generation phase of compiler and enrich the knowledge in code optimization techniques, code generation mechanism to construct and organize a compiler.	40

Mapping of CLO-PLO:

#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO1	Understand the knowledge of language translation to analyze the source program for compiler construction with its front-end and back-end phases.	PLO1	Cognitive/ Understand	Lecture, Class discussion.	Class test, Assignment, Exam
CLO2	Apply the theory of computing (patterns, tokens, regular	PLO2	Cognitive/Apply	Lecture, discussion,	Assignment Class test,



#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
	expressions, grammar and finite automata) to design and create the compiler for different programming languages and construction of compiler to improve language translation problem solving skills to basic real world situations			Problem solving.	Exam
CLO3	Apply the Intermediate code generation tools to design the intermediate code generation phase of compiler and enrich the knowledge in code optimization techniques, code generation mechanism to construct and organize a compiler.	PLO2	Cognitive/Apply	Lecture, Class discussion, Problem solving.	Class test, Assignment Exam

Course Content:

Segment	Contents	Duration	CLOs
Section-A (Midterm Exam: 30 Marks)			
1	Structure of compiler: Translator, Basic principles of compilers and compiler design, types of compilers, applications, phases of a compiler.	02	CLO1
2	Grammars & expressions: Concepts of languages and Grammars, Terminals and non-terminals, Useless non-terminals and Inaccessible tokens, Context-Sensitive Grammars, Context-Free Grammars, Right linear grammar, left linear grammar, Derivations and Parse trees, Ambiguous grammar.	05	CLO2
3	Automata: Lexical analysis, Regular expressions, finite automata, Non-deterministic Finite-State Automata, Minimization of Finite-State Automata, DFA, Symbol table management.	05	CLO2
Section-B (SEE: 50 Marks)			
4	Syntax Analysis: Parser, Error-Recovery, Left recursion, Left factoring, Top-Down Parsers, FIRST and FOLLOW, Recursive-Descent and Predictive Parsing, LL(1) parsing.	05	CLO3
5	Parsing: Bottom-Up Parsing, Operator Precedence parsing, LR Parsers, Construction of SLR, CLR, LALR Parse Tables, Shift-Reduce Parsing.	05	CLO3
6	Syntax-Directed Translation: Syntax-Directed definitions, Semantic Rules, Dependency Graph, Attributes, Construction of Syntax trees, Type checking, Type expressions, Translation Scheme, Type checking of expressions, statements and functions, Type	03	CLO3



	conversions.		
7	Intermediate code generation: Run time environments, Activation tree, Control stack, Run time storage organization, Storage allocation strategies, Intermediate languages, three address statements, quadruples, triples, indirect triples and polish notations.	02	CLO3
8	Code Generation and Optimization: Code generation issues, Instruction costs, Basic blocks, Flow graphs, DAG representation, Next use information, Transformations of Basic Blocks, Peephole optimization, Function preserving transformations, Optimization of Basic Blocks, Different models of code generation, error detection and error handling.	03	CLO3
		30	

Books :**Text Book :**

A. V. Aho, R. Sethi, J.D. Ullman Compiler Principle Techniques and Tools 2nd edition Pearson education 2007 ISBN- 81-7808-046-X

Reference Books :

1. William A Barret, R. M. Bates **Compiler Construction theory and practice** 2nd Edition University Grants Commission of Bangladesh 2006 ISBN-13: 978-9848090244
2. Alfred V Aho, Monica S Lam, Ravi Sethi, Jeffrey D. Ullman **Compiler principles, technique and tools** 2nd Edition Pearson 2007 ISBN-13 : 978-0321486813
3. John E. Hopcroft, Rajeev Matwani, Jeffrey D. Ullman **Introduction to Automata Theory, Language, and Computation** 2nd Edition 2005 Pearson ISBN-13 : 978-0201441246
4. A.J Holub **Compiler Design in C** Prentice-Hall 1990 ISBN-13 : 978-0131550452
5. Levine **Flex and Bison: Text Processing Tools** O'Reilly Media 2009 ISBN-0596155972

Course Assessment Pattern (Theory courses):

Bloom's Category		Evaluations out of 100 marks			
		CIE (50 marks)			SEE (50marks)
Cognitive learning	Affective learning	Mid-term (30)	Assignment/ Class Test (10)	Attendance Marks (10)	Written Exam (50)
Remember		11			10
Understand		4	10		10
Apply		13			30
Analyze		2			
Evaluation					
Create					
x	Responding	x	x	10	

Note: CIE=Continuous Internal Evaluation, SEE= Semester End Examination



5 th Semester		
ISCED Code	Course Code	Course Title
0613	CSE-3528	Compiler Lab
Credit Hours: 1	Contact Hours: 2	Type: Core, Engineering
Prerequisite: CSE-2427 (Theory of Computation)		
Co-requisite: CSE-3527 (Compiler)		

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Mid term	30 Marks
		Final term	30 Marks
	SEE: Semester End Examination		30 Marks

Objectives:

- This course is concerned with the practical aspects of designs based on the course CSE-3527, Compiler.

Course Learning Outcomes (CLOs):

#	CLO Description	Weightage (%)
1.	Implement Lexical Analyzer.	20
2.	Implement parsing algorithm and different types of parser.	50
3.	Construct a simple compiler.	30

Mapping of CLO-PLO:

#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO1	Implement Lexical Analyzer	PLO1	Cognitive/Apply	Lecture, Class discussion, Lab work.	Exam, Class performance, Assignment
CLO2	Implement parsing algorithm and different types of parser.	PLO2	Cognitive/Apply	Lecture, Class discussion, Lab work.	Assignment, Class performance
CLO3	Construct a simple compiler.	PLO2	Cognitive/Apply	Class discussion, Lab work.	Exam, Class performance, Assignment

Lecture Plan:

Week	Activities	Topics	CLOs
1	Lab Work	Introduction to compiler sessional, basic C programming structures, Basic compiler construction tools (LEX, YACC etc.)	CLO1
2	Lab Work	Implementation of comments remover of C program in C programming language	CLO1
3	Lab Work	Implementation of identifier and number checker of C program in C	CLO1
4	Lab Work	Implementation of tokenizer by using C programming language	CLO1
5	Lab Work	Implementation of Lexical analyzer using by LEX tool	CLO1
6	Lab Work	Implementation of Syntax analyzer using by YACC tool	CLO2



Week	Activities	Topics	CLOs
7	Lab Test		
8	Lab Work	Implementation of FIRST and FOLLOW of non-terminals from a CFG grammar	CLO2
9	Lab Work	Implementation of left factoring and Left recursion remover in C	CLO2
10	Lab Work	Implementation of Predictive Parsing Table using FIRST and FOLLOW from a grammar	CLO2
11	Lab Work	Implementation of Shift-reduce Parser from a CFG	CLO2
12	Lab Work	Construction of Calculator using LEX and YACC (Project)	CLO3
13	Lab Work	Implementation of intermediate code generation using C programming	CLO3
14	Lab Test	Lab report and project submission. Preparation for semester final examination	

Books :**Text Book :**

A. V. Aho, R. Sethi, J.D. Ullman Compiler Principle Techniques and Tools 2nd edition Pearson education 2007 ISBN-81-7808-046-X

Reference Books :

1. William A Barret, R. M. Bates Compiler Construction theory and practice 2nd Edition University Grants Commission of Bangladesh 2006 ISBN-13: 978-9848090244
2. Alfred V Aho, Monica S Lam, Ravi Sethi, Jeffrey D. Ullman Compiler principles, technique and tools 2nd Edition Pearson 2007 ISBN-13 : 978-0321486813
3. John E. Hopcroft, Rajeev Matwani, Jeffrey D. Ullman Introduction to Automata Theory, Language, and Computation 2nd Edition 2005 Pearson ISBN-13 : 978-0201441246
4. A.J Holub Compiler Design in C Prentice-Hall 1990 ISBN-13 : 978-0131550452
5. Levine *Flex and Bison: Text Processing Tools* O'Reilly Media 2009 ISBN-0596155972

**ISCED: 0613****Course Code: CSE-3529****Course Title: System Analysis and Design****Credit Hours: 3****Contact Hours: 3 lecture hours per week****Type: Core, Engineering****Prerequisite: N/A****Co-requisite: N/A**

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30Marks
	SEE: Semester End Examination		50 Marks

Course Objectives:

The student who completes this course should know:

1. The traditional systems/software development methodologies.
2. Traditional analysis and design techniques: entity-relationship diagrams, data flow diagrams, and data dictionaries.
3. How to use prototyping in the analysis and design phases of systems development.
4. Project planning: defining the scope, purpose and activities of a project; setting up a project management web page.
5. PERT/CPM project management techniques, including the use of Microsoft Visio.
6. What a CASE tool is and how it is used in software development.
7. How Object Oriented methodology techniques work.

Course Learning Outcomes (CLOs):

After completion of this course, students will be able to:

#	CLO Description	Weightage (%)
1	Understand project management and system development life cycle.	20%
2	Analyze requirement and feasibility of software application	20%
3	Design an information system.	40%
4	Compile the system design document	20%

Mapping of CLO-PLO:

#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO1	Understand project management and system development life cycle.	PLO1	Cognitive/Understand	Lecture, Class discussion, Assignment, Note	Exam, Quiz, Assignment



#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO2	Analyze requirement and feasibility of software application	PLO2	Cognitive/Analyze	Lecture, Class discussion, Assignment, Note	Exam, Quiz, Assignment
CLO3	Design an information system.	PLO2	Cognitive/Create	Lecture, Class discussion, Assignment, Lab work, Note, Complex Problem Solving	Exam, Quiz, Assignment,
CLO4	Compile the system design document	PLO2 PLO8 PLO9 PLO10	Cognitive/Create	discussion, Assignment	Exam, Assignment Report, Presentation

8 Stanza Syllabus:

Stanza	Topics
1	Concepts of system and its environment: Information, Types of information, Quality of information, System, Types of systems, Components of system, Source of information.
2	Information gathering: strategy, Information searching methods, Interviewing technique, System development methodologies and life cycle.
3	Feasibility study & Cost/Benefit analysis: Feasibility considerations, steps in feasibility analysis, feasibility report, Cost and Benefit categories, procedure for cost and benefit determination, classification of cost and benefit, cost and benefit evaluation methods.
4	Tools of analysis and design: Data Flow Diagram (DFD), DFD symbols, Constructing DFD; Data Dictionary; Decision Tree, Structured English, Decision Tables.
5	System Design and Construction: The process of design, System design phases, Design methodologies; Structured design; Form-Driven methodology; Input design, Output design, File and database design
6	Testing and Quality Assurance: Testing, Types of system tests; White-Box testing; Black-box testing; Quality factors specifications.
7	Implementation and Maintenance: Types of implementation, Documenting the system, Training and supporting users, Factor models of implementation success; The process of maintaining information system, Types of maintenance, Cost of maintenance, Reducing maintenance cost.
8	Hardware/ Software selection, control and security: Phases in selection, Criteria for software selection, Hardware selection, Financial considerations in selection; Security definitions, Threats to system security, Control measures,



	system failures and recovery.
--	-------------------------------

Text Books:

1. "Jeffrey Whitten Lonnie D. Bentley" Systems Analysis and Design Methods 7th Edition "McGraw-Hill Education" 2007 "ISBN: 0073052337, 9780073052335"
2. "Kenneth E. Kendall Julie E. Kendall" Systems Analysis and Design 8th Edition Prentice Hall 2010 ISBN-13: 978-0136089162

Reference Books:

1. "Alan Dennis Barbara Haley Wixom Roberta Marie Roth" Systems Analysis and Design, 5th Wiley 2012 "ISBN- 1118057627, 9781118057629"
2. "Akhilesh Bajaj" Systems Analysis and Design for Advanced Modeling Methods: Best Practices 1st Edition IGI Global 2009 ISBN-978-1-60566-344-9 Stanislaw Wrycza
3. "Alan Dennis Barbara Haley Wixom Roberta M. Roth" Systems Analysis and Design 4th Edition "John Wiley & Sons Inc." 2008 ISBN-13: 978-0137689958" Scott Tilley Harry J. Rosenblatt" Systems Analysis and Design 11th Edition Cwngage Learning 2017 ISBN-978-1-305-49460

Course Assessment Pattern

Bloom's Category		Evaluations out of 100 marks			
		CIE (50 marks)			SEE (50marks)
Cognitive learning	Affective learning	Mid-term (30)	Assignment/ Class Test (10)	Attendance Marks (10)	Written Exam (50)
Remember					
Understand		5	5		15
Apply					
Analyze		10	5		15
Evaluation					
Create		15			20
x	Responding	x	x	10	

Note: CIE=Continuous Internal Evaluation, SEE= Semester End Examination

Delivery methods & activities: Lecture, White Board Writing, Questions and Answers, Discussions Power point Presentation,

Assessment tools: Class Attendance, Class test, Quizzes/ Assignment on problem solution, Mid-Term & Final Exam. Project evaluation & Viva



ISCED: 0612

Course Code: CSE-3532

Course Title: Tools and Technologies for Internet Programming

Credit Hours: 2

Contact Hours: 4 lecture hours per week

Type: Core, Engineering

Prerequisite: Object Oriented Programming II (CSE 2401)

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30Marks
	SEE: Semester End Examination		50 Marks

Course Objectives:

- To understand the concept of Web Application Development and its Architecture.
- To understand the Essentials of Web Application Development.
- To understand and practice web page designing techniques.
- To understand and practice embedded dynamic scripting on client side Internet Programming.
- To understand the differences between client side & server side technologies to develop Web Application.

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to:

#	CLO Description	Weightage (%)
1.	Understand modern protocols and systems used on the Web (such as HTML, HTTP, URLs, CSS, XHTML, and XML).	10%
2.	Apply appropriate tools for client side and server side programming to make interactive web application.	10%
3.	Analyze the security issues relevant to web applications.	30%
4.	Develop, deploy, and maintain web application	40%
5.	Evaluate professional, ethical, environmental, and social impacts and responsibilities of the designed project.	10%

Mapping of CLO-PLO:

#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
---	-----------------	------	-------------------------------	---------------------------------	------------------



#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO1	Understand modern protocols and systems used on the Web (such as HTML, HTTP, URLs, CSS, XHTML, and XML).	PLO1	Cognitive/Understand	Lecture, Class discussion, Assignment, Note	Exam, Quiz, Assignment
CLO2	Apply appropriate tools for client side and server side programming to make interactive web application.	PLO1	Cognitive/Apply	Lecture, Class discussion, Assignment, Note	Exam, Quiz, Assignment
CLO3	Analyze the security issues relevant to web applications.	PLO1	Cognitive/Analyze	Lecture, Class discussion, Assignment, Lab work, Note	Exam, Quiz, Assignment
CLO4	Develop, deploy, and maintain web application	PLO2 PLO5 PLO3 PLO4 PLO9 PLO10 PLO11	Cognitive/ Create	Lecture, Class discussion, Assignment, Note	Exam, Quiz, Assignment
CLO5	Evaluate professional, ethical, environmental, and social impacts and responsibilities of the designed project.	PLO6 PLO7 PLO10	Cognitive/Evaluate	Lecture, Discussion	Presentation, Report

List of Experiments:

1. HTML: Basic, Elements, Attributes, Headings, Paragraphs, Formatting, Links, Head, Images, Tables, Lists, Blocks, Layout, Forms, Colors, Colornames, CSS, JavaScript, Entities, URL Encode, Quick List
2. CSS: Introduction, Syntax, Id & Class, Styling Backgrounds, Text, Fonts, Links, Lists, Tables, Box Model, Border, Outline, Margin, Padding, Grouping/Nesting, Dimension, Display, Positioning, Floating, Align, Navigation Bar, Image Gallery



3. JavaScript: Introduction, Output, Statements, Comments, Variables, Data Types, Objects, Functions, Operators, Comparisons, Conditions, Loop, Errors, DOM Introduction, Number, String, Date, Array, Window, Screen, Location, History, Navigator, Popup Alert
4. PHP part 1: Syntax, Variables, String, Operators, If...Else, Switch, Arrays, While Loops, For Loops, Functions, Forms, \$_GET, \$_POST
5. PHP part 2: Date, Include, File Upload, Cookies, Sessions, E-mail, Error, Exception, PHP Database, PHP ODBC
6. MySQL part 1: Select, Distinct, Where, And & Or, Order By, Insert, Update, Delete, Like, In, Between, Alias, Joins, Union, Create DB, Create Table, Primary Key
7. MySQL part 2: Foreign Key, Drop, Alter, Increment, Views, Dates, Functions- avg(), count(), max(), Group By, Having, format()

Text Books:

1. "Paul Deitel" Internet & World Wide Web How to Program, Fifth Edition "Pearson Education Limited England" 2011 ISBN-10: Harvey Deitel and Abbey Deitel, ISBN-10: 0132151006, ISBN-13: 9780132151009

Reference Books:

1. Marty Hall and Core Web programming, Second Edition "Prentice Hall USA" 2001 ISBN: 987130897930 ISBN-13: 978-0130897930
2. Jeffrey C. Jackson Web Technologies: A Computer Science Perspective First Edition "Pearson Education Limited England" 2006 ISBN: 987131856030 ISBN-13: 978-0131856035
3. Patel Bankim and Bihari Barik Lal Introduction to Web Technology and Internet First Edition "Acme Learning Private Limited New Delhi" 2009 ISBN: 9878190710435 ISBN-13: 978-8190710435
4. Alexis Leon and Mathews Leon Internet for Everyone Second Edition "Vikas Publishing House" 2012 ISBN-10: New Delh, ISBN-13: 978-8182093188



ISCED: 0714

Course Code: EEE-2421

Course Title: Electrical Drives and Instrumentation

Credit Hours: 2

Contact Hours: 2 lecture hours per week

Type: Core, Engineering

Prerequisite: EEE-1121 (Basic Electrical Engineering)

Co-requisite: EEE-2422 (Electrical Drives and Instrumentation Sessional)

This course aims to equip students with the knowledge of transformer, DC and AC electrical drive systems which are commonly used in industries. The course also covers the general concepts of electrical measurement and instrumentation techniques.

Objectives:

This course is designed to provide students. The material will be presented using the normal mix of lectures and laboratory experiments and demonstrations. This course objective is also to

- To provide technical knowledge to solve industrial problems related to AC and DC Drives systems
- To introduce the students to electrical measurement and instrumentation techniques.
- To introduce the basic concepts and working principles of electrical machines

Course Learning Outcomes (CLOs):

At the completion of the subject, students should be able to perform the following tasks:

#	Course Learning Outcomes	Weightage (%)
1.	Understand the operation of transformer, DC /AC Drives for industrial applications.	25
2.	Solve the problems in AC/DC Variable Speed Drive system used for the given industrial application.	20
3.	Select an appropriate motor for a given industrial application.	20
4.	Understand various measuring techniques for both electrical and non-electrical quantities.	25
5.	Comprehend the fundamentals and differentiate errors of an instrumentation measurement system.	10

Mapping of CLO-PLO:

#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO1	Understand the operation of transformer, DC /AC Drives for industrial applications.	PLO1	Cognitive/understand	Lecture, Class discussion, Assignment, Note	Exam, Quiz, Assignment



#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO2	Solve the problems in AC/DC Variable Speed Drive system used for the given industrial application.	PLO2	Cognitive /apply	Lecture, Class discussion, Assignment, Note	Exam, Quiz, Assignment
CLO3	Select an appropriate motor for a given industrial application.	PLO2	Cognitive / apply	Lecture, Class discussion, Assignment, Note	Exam, Quiz, Assignment
CLO4	Understand various measuring techniques for both electrical and non-electrical quantities.	PLO1	Cognitive/understand	Lecture, Class discussion, Assignment, Note,	Exam, Quiz, Assignment
C05	Comprehend the fundamentals and differentiate errors of an instrumentation measurement system.	PLO2	Cognitive/analyze	Lecture, Class discussion, Assignment, Note	Exam, Quiz, Assignment

Course Content:**Section-A (Mid-term: 30 Marks)****1. Introduction to Electrical Drives**

Introduction to electrical machines. Rotational motion, Newton's law, and power relationships. Magnetic field, Faraday's law, induced voltage on a conductor moving in a magnetic field, production of force on a wire in a magnetic field, Classification of Load torques, Mechanical and Electrical Power Calculation, Sizing of electric motors for given load system, Classes of Motor Duty, De-rating factor for electric motor sizing, Energy Efficient Motors, Motor name plate.

2. Transformers: Single-phase transformers: Construction, principle of operation and equivalent circuit, phasor diagram, efficiency and regulation. Short and open circuit tests. Three-phase transformers: Construction and connections.

3. DC Motor Drives: Principles of operation and construction of DC machines, Emf equation and principle of commutation. Controlled rectifier fed dc drives, Power factor, supply harmonics and ripple in motor current, Chopper controlled dc drives, Closed loop control of DC Drives, Two and four quadrant controls.

Section-B (Final Exam: 50 Marks)**Group-A (20 Marks)**

4. Induction Motor Drives: Principles of operation and basic construction of three-phase induction motor. Slip equation, equivalent circuit, determination of equivalent circuit parameters by no-load and blocked-rotor tests. Volts per Hertz drives, Flux vector control drives, Direct torque control drives, Soft-Starters, Selection of speed drives and soft starters, Line reactors, Troubleshooting of AC Drives system, Drives parameter programming.



5. **Synchronous Motor Drives:** Stepper Motor Drives, Switched reluctance motor drives, Permanent Magnet Synchronous Motor (Interior Permanent Magnet and Surface Permanent Magnet, Brushless DC motor). Thyristor and microprocessor based speed control of motors.

Group-B (30 Marks)

6. **General Concepts of Measurements and instrumentations**
Variables and measurement signals, the three stages of generalized measurement system, some common terms used in the measurement system, mechanical loading, impedance matching, and frequency response. Factors considered in selection of instruments - Measurement accuracy and precision. Error analysis and classification, sources of error.
7. **Instrumentation amplifiers:** Differential, logarithmic and chopper amplifiers; Frequency and voltage measurements using digital techniques; Recorders and display devices, spectrum analyzers and logic analyzers
8. **Measurement of Non Electrical Quantities: Transducers-** terminology, types, principles and application of photovoltaic, piezoelectric, thermoelectric, variable reactance and optoelectronic transducers; Measurement of Temperature: Resistance thermometer, and thermo couples.

Text Books:

1. Malaric, R., "Instrumentation and Measurement in Electrical Engineering", Brown Walker Press, 2011.
2. Stephen J. Chapman : Electric Machinery Fundamentals

Reference Books:

3. Sawhney, A.K., "A Course in Electrical and Electronic Measurement and Instrumentation", Dhanpat Rai & Sons, New Delhi, 2001
4. Ernest, O. Doebelin., "Measurement Systems: Applications and design", Mc-Graw Hill, 2004.
5. Beckwith, T.G and Buck, N.L., "Mechanical Measurements", Addition Wesley, 2007.
6. Golding, E.W., and Widdis, F.C., "Electrical Measurements and Measuring Instruments", A H Wheeler & Company, Calcutta, 1993.
7. B.L. Theraja & A.K. Theraja : A textbook of Electrical Technology

Course Assessment Pattern (Theory courses):

Bloom's Category		Evaluations out of 100 marks			
		CIE (50 marks)			SEE (50marks)
Cognitive learning	Affective learning	Mid-term (30)	Assignment/ Class Test (10)	Attendance Marks (10)	Written Exam (50)
Remember	-		-	-	5
Understand	-	5	5	-	15
Apply	-	10	-	-	20
Analyze	-	15	5	-	10
Evaluation	-			-	
Create	-		-	-	
x	Responding	x	x	10	

Note: CIE=Continuous Internal Evaluation, SEE= Semester End Examination

Delivery methods & activities: Lecture, White Board Writing, Questions and Answers, Discussions Power point Presentation,

Assessment tools: Class Attendance, Class test, Quizzes/ Assignment on problem solution, Mid-Term & Final Exam. Project evaluation & Viva



ISCED: 0714

Course Code: EEE-2422

Course Title: Electrical Drives and Instrumentation Sessional

Credit Hours: 2

Contact Hours: 2 lecture hours per week

Type: Core, Engineering

Prerequisite: EEE-1121 (Basic Electrical Engineering)

Co-requisite: EEE-2422 (Electrical Drives and Instrumentation)

Course Learning Outcomes (CLOs):

At the completion of the subject, students should be able to perform the following tasks:

#	Course Learning Outcomes	Weightage (%)
1.	Apply engineering knowledge and follow instructions to perform experiments in (i) Electrical drives and (ii) Measurement and Instruments	20
2.	Perform the experimental using appropriate engineering tools	30
3.	Develop a project based on achieved theoretical knowledge	30
4.	Comply with ethics values and responsibility towards others in safety and bench tidiness	20

Mapping of CLO-PLO:

#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO1	Apply engineering knowledge and follow instructions to perform experiments in (i) Electrical drives and (ii) Measurement and Instruments	PLO1	Cognitive/Apply	Lecture, Class discussion, Assignment, Note	Pre-Lab Logbook, Lab Report, Lab Performance, Open Ended
CLO2	Perform the experimental using appropriate engineering tools	PLO5	Cognitive/Applying	Lecture, Class discussion, Assignment, Note	Lab Performance, Lab Report,
CLO3	Develop a project based on achieved theoretical knowledge	PLO3 PLO9 PLO10 PLO11	Cognitive/Create	Lecture, Class discussion, Assignment, Lab work, Note	Project, Report



#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO4	Comply with ethics values and responsibility towards others in safety and bench tidiness	PLO8 PLO10	Cognitive/Analyze	Lecture, Class discussion, Assignment, Lab work, Note	Lab Performance, Lab report

Experiment Names:

1. To study the characteristics of motor
2. To study the characteristics of chopper amplifier
3. To construct a multimeter.
4. To perform also other experiments relevant to this course.

Text Books:

1. Malaric, R., "Instrumentation and Measurement in Electrical Engineering", Brown Walker Press, 2011.
2. Stephen J. Chapman : Electric Machinery Fundamentals

Reference Books:

3. Sawhney, A.K., "A Course in Electrical and Electronic Measurement and Instrumentation", Dhanpat Rai & Sons, New Delhi, 2001
4. Ernest, O. Doebelin., "Measurement Systems: Applications and design", Mc-Graw Hill, 2004.
5. Beckwith, T.G and Buck, N.L., "Mechanical Measurements", Addition Wesley, 2007.
6. Golding, E.W., and Widdis, F.C., "Electrical Measurements and Measuring Instruments", A H Wheeler & Company, Calcutta, 1993.
7. B.L. Theraja & A.K. Theraja : A textbook of Electrical Technology



Cognitive learning	Affective Learning	Mid-term: (30)	Assignment/ Class Test: (10)	Attendance Marks (:10)	Written Exam: (50)
Remember	-	-	-	-	5
Understand	-	5	5	-	5
Apply	-	5	-	-	10
Analyze	-	5	-	-	10
Evaluation	-	10	5	-	10
Create	-	5	-	-	10
x	Responding	x	x	10	
Remarks	Course teachers may change the magnitude of marks in Bloom's category(Both for CIE and SEE), but he/she will have to keep in mind that the % of higher order learning mode must be about 60% or more and all the Bloom's categories to be addressed during the semester.				

Note: CIE=Continuous Internal Evaluation, SEE= Semester End Examination.

Delivery methods & activities: Lecture, White Board Writing, Questions and Answers, Discussions Power point Presentation,

Assessment tools: Class Attendance, Class test, Quizzes/ Assignment. Mid-Term & Final Exam. Project evaluation & Viva.

Course Code: URED 3503

Course Title: Political Thoughts and Social Behavior

Credits: 1 CH

Contacts: 2 CH Per Week

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30Marks
	SEE: Semester End Examination		50 Marks

The Rationale of the Course: Following are the Objectives of the course:

- To help the students demonstrate an in-depth understanding of their real position in this temporary world.
- To understand their relation with the state and society.
- To help them in performing their duties and responsibilities towards their state, society and humanity at large.

SL	Course Learning Outcomes (CLOs) : Upon the successful completion of the course, students will be able to :	Corresponding THE UNIVERSITY MS	Bloom's taxonomy domain/level
CLO1	Acquire basic knowledge about politics, Islamic politics, Law & Constitution, State and Government	THE UNIVERSITY MS-1,2&3	Cognitive/Understand
CLO2	Realize the nature of Citizen and Citizenship and relationship between state and citizen, citizen and government as well	THE UNIVERSITY MS-1,2&3	Cognitive/Apply
CLO3	Understand about the dealings and behavior of a man relating to himself, his Family and society	THE UNIVERSITY MS-1,2&3	Cognitive/Understand
CLO4	Analyze the principles of permissible and prohibition in Islam in regards to dress, food and drinking and in other parts of life	THE UNIVERSITY MS-1,2&3	Cognitive/Analyze
CLO5	Evaluate the Economic System of Islam and Festivals in Islam	THE UNIVERSITY MS-1,2&3	Cognitive/Evaluate

Course Contents:

Chapter	Chapter Title	Contents	Number of Lectures	Corresponding CLOs
Department of Computer Science and Engineering (CSE)				



Section A: Mid Term 30 Marks

Politics and Islamic Politics:

- | | | | |
|----|---|---|------|
| | A. Nature of the course from viewpoint of Islamic jurisprudence | | |
| 01 | B. Literal and terminological Meaning of of Politics | 3 | CLO1 |
| | C. Definition of Islamic Politics | | |
| | D. Basic Principles of Islamic Political System | | |
| | E. Conventional Political System and Islam | | |

Shariah and constitution :

- | | | | |
|----|--|---|------|
| | A. Meaning of Shariah | | |
| 02 | B. Sources of Shariah | 3 | CLO1 |
| | C. Difference between Shariah and man-made law | | |
| | D. Definition of Constitution | | |
| | E. Contents and main features of an Islamic Constitution | | |

State and Government:

- | | | | |
|----|--|---|------|
| | A. Meaning of State | | |
| | B. Elements of State | | |
| | C. The major characteristics of modern national states | | |
| 03 | D. Meaning of Islamic State | 3 | CLO1 |
| | E. Basic principles of an Islamic state | | |
| | F. Meaning, Objective, types of government | | |
| | G. Relation between state and government | | |
| | H. Organs of a government - Executive, Legislative and Judiciary | | |

Citizen and Citizenship:

- | | | | |
|----|--------------------------------|---|------|
| | A. Meaning of citizenship | | |
| 04 | B. Types of citizenship and | 3 | CLO2 |
| | C. Rights & duties of citizens | | |

Section B-

Semester End Examination (SEE): 50 Marks

Marriage and Family:

- | | | | |
|----|---|---|------|
| | A. Marriage in Islam | | |
| | B. Objective, Nature and some features of marriage in Islam | | |
| 05 | C. Classification of Marriage | 3 | CLO3 |
| | D. Legal effects of various types of marriage | | |
| | E. Prohibited Marriages | | |
| | F. The general essentials of a Muslim marriage | | |

Matrimonial Disputes Resolution and Divorce:

- | | | | |
|----|---|---|------|
| | A. Definition of Matrimonial Disputes | | |
| 06 | B. Sources and causes of matrimonial disputes | 3 | CLO3 |
| | C. Quranic Approach to matrimonial disputes | | |
| | D. Divorce as a last resorts to dissolve a matrimonial disputes | | |

Women in Islam:

- | | | | |
|----|--|---|------|
| 07 | A. Status of women in Islam | 2 | CLO3 |
| | B. Rights and duties of women in Islam | | |

Family Relationship Social Relationship & Festivals in Islam:

- | | | | |
|----|---|---|------|
| | A. Husband-wife relationship | | |
| | B. Parent-child relationship | | |
| 08 | C. Family relationship | 3 | CLO3 |
| | D. Kinship | | |
| | E. Neighbourhood | | |
| | F. Wide circle of relationship relationship | | |
| | G. Eid-al-Fitr and Eid-Al-Adha | | |

Economic System & Dress Code in Islam:

- | | | | |
|----|--|---|------|
| 09 | A. Earning and Expenditure by Halal Means | 3 | CLO5 |
| | B. Right to property and Individual Liberty | | |
| | C. System of <i>Zakah</i> (Welfare Contribution) | | |



- D. Prohibition of *Riba* (Interest or usury)
- E. Law of Inheritance (*Mirath*) , Conclusion
- F. Common Clothing Regulations
- G. Clothing Regulations for Man
- H. Clothing Regulations for Woman

Principles of permissible and prohibition:

- A. All things are lawful for mankind except what have been prohibited explicitly by Allah and His Messenger
- B. Only Allah (SWT) has the right to declare a thing lawful or prohibited
- 10 C. Allah has prohibited only shameful things 4 CLO4
- D. What has been made lawful for us is sufficient
- E. the means or causes that lead to unlawful action are also prohibited
- F. what is prohibited, it is prohibited for all
- G. a grave necessity legalizes temporarily an illegal thing

References:

1. Bhuiyan, Mohammad Shafiul Alam (2007), *The Government and Politics in Islam*, Dhaka: Noor Publications.
2. Hamid, E. A. (2004). *The Qur'an and Politics*, London: International Institute of Islamic Thought.
3. Islam, A. B. M. M. (2005). *Islamic Constitution: Quranic and Sunnatic Perspectives*.
4. A'LaMaududi, S. A. (1980). *The Islamic Law & Constitution*: Islamic Books.
5. Sarwar, G. (1982). *Islam, beliefs and teachings*: Muslim educational trust.
6. Hannan, S. A. (2017). *Social Laws of Islam*. Dhaka: Bangladesh Institute of Islamic Thought (BIIT)
7. Zino, Muhammad bin Jamil, *Islamic Guidelines*, Darusalam, Riyadh, 1996.
8. Al-Qaradawi, Y. (1988). *Halal and Haram in Islam*.
9. Ahmad, M. (2009). *Business ethics in Islam*: International Institute of Islamic Thought (IIIT).
10. Chapra, M. U. (2016). *The future of economics: An Islamic perspective* (Vol. 21): Kube Publishing Ltd.

Course Assessment Pattern (Theory courses):

Bloom's Category		Evaluations out of 100 marks			
		CIE (50 marks)			SEE (50marks)
Cognitive learning	Affective Learning	Mid-term: (30)	Assignment/ Class Test: (10)	Attendance Marks (:10)	Written Exam: (50)
Remember	-	5	-	-	5
Understand	-	-	5	-	10
Apply	-	5	-	-	05
Analyze	-	5	-	-	10
Evaluation	-	10	5	-	15
Create	-	5	-	-	05
x	Responding	x	x	10	
Remarks	Course teachers may change the magnitude of marks in Bloom's category(Both for CIE and SEE), but he/she will have to keep in mind that the % of higher order learning mode must be about 60% or more and all the Bloom's categories to be addressed during the semester. If necessary, a course teacher may also use Cognitive (Knowledge), Affective (Attitude) and Psychomotor (Skills) domain of Bloom's Taxonomy.				

Note: CIE=Continuous Internal Evaluation, SEE= Semester End Examination.

- Delivery methods & activities:** Lecture, White Board Writing, Questions and Answers, Discussions Power point Presentation,
- Assessment tools:** Class Attendance, Class test, Quizzes/ Assignment.Mid-Term & Final Exam. Project evaluation & Viva

Course Code: URED 3604

Course Title: The Life and Teachings of the Prophet Muhammad (SAAS)

Credits: 1 CH

Contacts: 2 CH